

Question: 1

Research two actual network switches available in India (for example, from Cisco, TP-Link, or D-Link) and list their key specifications such as number of ports, speed, and price in a table.

Answer: 1

Comparison of Actual Network Switches Available in India

Two highly reliable entry-level unmanaged network switches widely used for home offices, small setups, and commercial network layouts in India are compared below:

| Network Switch Brand & Model     | Total Number of Ports | Downlink Data Speeds       | Type / Management             | Average Online Market Price |
|----------------------------------|-----------------------|----------------------------|-------------------------------|-----------------------------|
| TP-Link TL-SG1024D               | 24 Ports (RJ45)       | 10/100/1000 Mbps (Gigabit) | Unmanaged Desktop / Rackmount | ₹5,505                      |
| <a href="#">D-Link DGS-1016D</a> | 16 Ports (RJ45)       | 10/100/1000 Mbps (Gigabit) | Unmanaged Desktop / Rackmount | ₹6,031                      |

Key Specification Insights

- TP-Link TL-SG1024D:** This [24-port Gigabit Switch](#) is designed for high-density wired networks. It is unmanaged, meaning it features plug-and-play setup requiring zero underlying software configuration. It features an energy-efficient design that cuts power automatically based on link status.
- D-Link DGS-1016D:** This [16-port Copper Gigabit Switch](#) offers a slightly more premium metal housing built for durability. It includes "D-Link Green" features that sense attached device power cycles and cable length to intelligently minimize operational power overhead.

Question: 2

Create a device selection matrix comparing at least one switch, one router, and one firewall that could be used for a small startup office (20-30 employees). Include columns for brand/model, key specs, price, and your rating for each device.<br><br><em><strong>Hint:</strong> Use online sources like Amazon India, Flipkart, or manufacturer websites for real-world pricing and specs.</em>

Answer: 2

Device Selection Matrix for a 20-30 Employee Startup

For a small office with **20 to 30 employees**, hardware reliability, price efficiency, and ease of management are essential factors. Below is a real-world device selection matrix comparing a compatible network switch, business router, and firewall solution available in India:

| Device Type       | Brand / Model             | Key Hardware Specifications                                                                                                                                                                         | Average Market Price (India) | Personal Quality Rating |
|-------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------|
| Network Switch    | Cisco Business 250-24T-4G | <ul style="list-style-type: none"><li>• 24 Gigabit RJ45 Ports</li><li>• 4x 1G SFP Uplink Slots</li><li>• Layer 2 Smart Managed, VLAN &amp; QoS Support</li></ul>                                    | ₹18,500                      | ★★★★☆<br>(4.5/5)        |
| Business Router   | TP-Link ER605 (Omada VPN) | <ul style="list-style-type: none"><li>• 1 Gigabit WAN Port + 3 WAN/LAN Changeable Ports</li><li>• High-speed IPSec/PPTP/L2TP VPN Support</li><li>• Centralized Cloud Management Interface</li></ul> | ₹4,499                       | ★★★★☆<br>(4.2/5)        |
| Hardware Firewall | Fortinet FortiGate 40F    | <ul style="list-style-type: none"><li>• 5x Gigabit Ports (WAN/LAN)</li><li>• 1 Gbps Threat Protection Throughput</li><li>• Advanced SSL Inspection &amp; Antivirus Web Filtering</li></ul>          | ₹36,200                      | ★★★★★<br>(4.9/5)        |

Hardware Role Justification

- **Cisco Business 250 Switch:** Provides enough physical ethernet ports to connect all 20–30 employee desk workstations, printers, and Wi-Fi access points directly to the local area network (LAN). Its smart web dashboard makes setting up internal department segregation (VLANs) simple.
- **TP-Link ER605 Business Router:** An extremely affordable gateway router capable of handling multi-ISP load balancing. If your primary internet line goes down, it can

instantly route traffic to a backup line, ensuring the office stays online with zero downtime.

- **Fortinet FortiGate 40F Firewall:** The most crucial piece for network safety. It sits directly between your internet modem and internal office devices, running deep packet inspection to block ransomware threats, malware, and unauthorized remote access attempts.

### Question: 3

Imagine you are setting up the network for a new gaming café with 25 computers and public Wi-Fi. Select one switch, one router, and one firewall model for this scenario and justify your choices in 2-3 lines each.

### Answer: 3

#### Gaming Café Device Selection Plan (25 PCs + Public Wi-Fi)

- **Network Switch: TP-Link TL-SG1024D (24-Port Gigabit) + Netgear GS108 (8-Port Gigabit)**
  - **Justification:** Combining these two unmanaged switches provides 30 total plug-and-play Gigabit ports, easily connecting all 25 gaming rigs and local server assets. This setup ensures zero latency drops or packet loss during high-speed local multiplayer matches without requiring complex software management.
- **Business Router: TP-Link ER605 (Omada VPN)**
  - **Justification:** This router features dedicated Multi-WAN load-balancing ports, allowing the gaming café to connect up to three separate internet connections at the same time. If one ISP drops, it instantly shifts connections to the backup line, preventing sudden disconnections for active gamers.
- **Hardware Firewall: Fortinet FortiGate 40F**
  - **Justification:** This hardware handles enterprise-grade security filtering natively without adding latency to the gaming rigs. It lets you create isolated network zones, ensuring that public Wi-Fi guests cannot access the main gaming PC network or local servers.

Question: 4

Prepare a Bill of Materials (BOM) listing the selected network devices (switch, router, firewall) for your gaming café scenario, including quantity, model, unit price, total price, and a short justification for each item.

Answer: 4

Bill of Materials (BOM) for Gaming Café Setup

| Item # | Device Category   | Brand & Model          | Qty | Unit Price (INR) | Total Price (INR) | Short Justification                                                                   |
|--------|-------------------|------------------------|-----|------------------|-------------------|---------------------------------------------------------------------------------------|
| 1      | Network Switch    | TP-Link TL-SG1024D     | 1   | ₹5,505           | ₹5,505            | Provides 24 dedicated Gigabit ports for direct, lag-free wired gaming rigs.           |
| 2      | Expansion Switch  | Netgear GS108          | 1   | ₹2,100           | ₹2,100            | Adds 8 extra Gigabit ports to cover the remaining gaming PCs and server assets.       |
| 3      | Business Router   | TP-Link ER605          | 1   | ₹4,499           | ₹4,499            | Handles Multi-WAN load-balancing to keep a backup internet line active for gamers.    |
| 4      | Hardware Firewall | Fortinet FortiGate 40F | 1   | ₹36,200          | ₹36,200           | Blocks external malware threats and isolates public Wi-Fi guests from gaming systems. |
| —      | —                 | —                      | —   | Total Budget     | ₹48,304           | Estimated total hardware cost for a 25-computer deployment.                           |

Question: 5

Perform a cost-benefit analysis for your chosen devices: explain in 3-4 lines why you selected these models over cheaper or more expensive alternatives, considering both features and budget.

Answer: 5

Cost-Benefit Analysis for Gaming Café Network

- **Why we skipped cheaper alternatives:** Going with cheaper home-use unmanaged routers or basic plastic switches would cause the network to crash or freeze under heavy gaming traffic. Cheaper devices cannot handle multi-ISP backup lines or keep

public Wi-Fi traffic separated from the main systems, leading to severe game lag and immediate security vulnerabilities for the entire café.

- **Why we skipped expensive alternatives:** Upgrading to high-end enterprise hardware (like Cisco Catalyst switches or high-capacity FortiGate 60F firewalls) would cost over ₹1.5 Lakhs. Those advanced models offer features like deep fiber-optic line routing or massive cloud database tracking, which are unnecessary overhead costs for a single-room setup running just 25 gaming computers.
- **Value balance of our selected models:** The chosen hardware bundle hits the perfect sweet spot by delivering essential commercial features—such as reliable **Gigabit wired speeds, Multi-WAN line backup security, and isolated network protection**—for a total budget of under ₹50,000. This configuration gives gamers a smooth, lag-free experience while keeping the initial business setup costs highly affordable.